One-pot PCR mixture and conditions (head-tailed gene-region primers + barcoded-head primers in a single reaction)

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Source of record: HuangLab (Yin-Tse Huang lab, NCKU). Reaction mixture and cycling program are the lab's working conditions from the sheet *ONE POT PCR.xlsx* (project `LIMS/protocol/raw_data`, 2026-10-20). Text, QC and LIMS steps follow the lab's two-step PCR protocol (*2-step PCR mixture and conditions, barcoded-head primers*). Companion workflow chart: *One-pot PCR amplicon library workflow* on the lab protocol page.

Abstract

Amplify a gene region and attach the Nanopore barcode in **one reaction**. The head-tailed gene-region primers are at low concentration and drive the first phase; the barcoded-head primer pair is in excess and takes over in the second phase by priming on the head sequence. The product is the same molecule the two-step route makes: *barcode – head – gene region – head – barcode*. Each sample gets its own tube with its own barcode; tubes are pooled equimolar afterwards. The conditions below are set up for **large amplicons (~3–4 kb)**; shorten the extension steps for smaller regions.

When to use one-pot instead of two-step

- Many samples, same gene region, templates that amplify well (clean fungal or bacterial DNA, or soil extracts that already work in the two-step route).
- You want to halve hands-on time and the number of opportunities for cross-contamination.
- Do **not** use one-pot for difficult templates (inhibitor-rich soil, low-copy targets) or when a gel-checkable intermediate is needed: there is no 1st-PCR product to inspect.

Materials

1. 2X KAPA HiFi HotStart ReadyMix (LIMS `taq` default: `KAPA HiFi HotStart`)
2. Head-tailed gene-region primers, forward + reverse, 10 µM working stocks (large-amplicon pair, ~4 kb; same primers as the 1st PCR of the two-step protocol — see the lab primer list for the region and its annealing temperature)
3. Pre-mixed barcoded-head primer pairs (forward + reverse), 10 µM each, one pair per barcode, from the barcode plate used for the 2nd PCR of the two-step protocol
4. DEPC-treated (nuclease-free) water
5. DNA template, normalised where possible (1–10 ng/µL); MolWork# assigned
6. 8-strip PCR tubes or 96-well PCR plate, filter tips, 1% bleach for the bench

NOTE Barcoded-head primers must carry the **same head sequence** as the tail of the gene-region primers; otherwise the second phase cannot prime. Check the primer list before mixing a new region with the barcode plate.

Reaction mixture (per 50 µL reaction)

Prepare the master mix **without barcode primers and template**, dispense, then add the barcode pair and the template per tube. Prepare 1.2X of the master mix volume for pipetting loss. **A negative control (no template) is always included**, carrying its own barcode.

Component	Volume	Final conc. (10 µM stocks)	Role
2X KAPA HiFi HotStart ReadyMix	25 µL	1X	polymerase + buffer + dNTPs
Gene-region primer F, head-tailed (10 µM)	0.5 µL	0.1 µM	phase 1
Gene-region primer R, head-tailed (10 µM)	0.5 µL	0.1 µM	phase 1
DEPC-treated water	19.5 µL	—	to 50 µL
Master mix subtotal	*45.5 µL*		
Pre-mixed barcoded-head primers F+R (10 µM each)	1.5 µL	0.3 µM each	phase 2
DNA template	3 µL		
Total	**50 µL**		

NOTE Volumes are for a 50 µL reaction (lab sheet). For a 25 µL reaction halve every volume. Keep the gene-region : barcode primer ratio (1 : 3) when scaling.

Procedure

Setup

1. Wear gloves. Clean the bench with 1% bleach. Thaw the ReadyMix, primers and templates on ice; vortex briefly and spin down.
2. Prepare the master mix (ReadyMix + gene-region primers + water) for n + 1 reactions × 1.2. Mix gently by pipetting; quick spin.
3. Dispense **45.5 µL** master mix into each PCR tube / well.
4. Add **1.5 µL** of the pre-mixed barcoded-head primer pair to each tube, **one barcode per tube**. Write the barcode ID on the tube and in the LIMS PCR row before continuing.
5. Add **3 µL** DNA template to each tube. The negative control receives 3 µL water instead.
6. Mix gently by tapping, quick spin. Keep on ice until the cycler is ready.

Cycling program (seven stages, single run)

7 Run the program below (lab sheet *ONE POT PCR.xlsx*, sheet *PCRcondition*). Phase 1 (stage 2) anneals at the **gene-region primer** temperature; phase 2 (stage 5) anneals at **65 °C**, where the barcoded-head primers prime on the head sequence.

Stage	Step	Temp	Time	Cycles	Purpose
1	Initial denaturation	98 °C	45 s	1×	
2	Denaturation	98 °C	10 s	**25×**	phase 1: gene region
2	Annealing	**60–66 °C** (a)	30 s		
2	Extension	72 °C	2 min 30 s (b)		suitable for 3–4 kb amplicons
3	Intermediate extension	72 °C	5 min	1×	completes phase-1 products
4	Denaturation	98 °C	30 s	1×	transition to phase 2
5	Denaturation	98 °C	15 s	**15×**	phase 2: barcoding
5	Annealing	65 °C	15 s		
5	Extension	72 °C	30 s (b)		
6	Final extension	72 °C	2 min 30 s	1×	
7	Hold	4 °C	∞		

(a) Choose the phase-1 annealing temperature from the gene-region primer list; the LIMS `method` value records it as `one_pot_60`, `one_pot_62` or `one_pot_64`. (b) Extension times are set for ~3–4 kb products; scale at ~1 min per kb for phase 1. The phase-2 extension of 30 s matches the two-step protocol's 2nd PCR, where the barcoded primers only need to copy across the head.

NOTE 40 cycles in total (25 + 15). Do not add cycles to phase 2: over-cycling increases chimeras and barcode-primer dimers. If the product is weak, check the template amount and the phase-1 annealing temperature first.

Quality control

8 Run **2 µL** of each reaction on a 1% agarose gel next to the ladder. Expected product = gene-region amplicon + ~2 × (head + barcode). One band per lane; the negative control must be blank.

Observation	Interpretation	Action
Single band at expected size	one-pot worked	continue
Two bands: expected + smaller (head-only)	phase 2 incomplete	check barcode pair was added; re-run phase 2 on the product if needed
Smear or band below 150 bp	primer dimer	reduce template / check primer stocks
Band in negative control	contamination	discard the batch, re-clean the bench

9 Photograph the gel, mark the wells, upload to the lab Google Drive, and record `pcr_ok` for each row.

10 Quantify each tube on the **Qubit Flex** (dsDNA HS). Record the concentration in the LIMS; this value drives the equal-molar pooling.

LIMS registration (Stage 3b, `/lims-register`)

11 One `pcr_reaction` row per MolWork#. Set `method` = `one_pot_<Ta>`, `taq` = `KAPA HiFi HotStart`, primers, date, performed_by. Because the barcode is already in the single reaction, assign the **flow cell and barcode at setup** and take the single OK: the grid mirrors `pcr_ok` → `second_pcr_ok` for `one_pot_*` rows, so the sample becomes ready for Stage 4a as soon as the gel passes.

Downstream

12 Pool all barcoded tubes **equal molar**, then purify the pool with the *Size selection (Purification) (1X ratio)* protocol and proceed to library construction (EvoPrep). See the one-pot workflow chart for the full path.

Change log

- v1.0 (2026-10-20): reaction mixture and cycling program taken from the lab sheet *ONE POT PCR.xlsx*; replaces the v0.1 draft values.